# WHAT DO WE KNOW?

CHAPTER 3

**"WHEN I WAS GROWING UP, EVERY HOUSE ON MY BLOCK WAS OCCUPIED. DURING LAST MONTH'S SURVEY, I SAW THE FIRST VACANT HOME ON MY STREET....I DON'T WANT MY HISTORY TO BE ERASED."**

**Lauren Hood, Community Engagement Manager at Loveland Technologies, Director of Deep Dive Detroit, and Participant in Motor City Mapping**

# BLIGHT'S IMPACTS

Blight is a drag on community energy. It is a siphon on city vitality. Blight is a strong deterrent to economic investment and a proven threat to public safety. Blight can be a source of despair or cynicism for people who have witnessed the decline of a particular building or neighborhood over time. Broken windows, piled-up trash, and scorched or stripped interiors of vacant structures have challenged Detroiters every day for decades. This has been a painful reality for the city's people and has harmed the national image of Detroit.

Eliminating Detroit's blight will dramatically improve the next chapter of the city's story, as community activists, faith leaders, designers, artists, greeners, block captains, business owners, individual residents, and organizations of every size and description implement a bold and better vision for the future.

*Neighbors in Brightmoor, which lost one-third of its population between 2000 and 2010, have worked energetically to capture the creativity and enthusiasm of residents. Working through the broad-based Brightmoor Alliance, residents have created farms, gardens, community and school art projects.*

*In nearby Cody Rouge, the Stahelin Block Association president, Andrea Jackson, gathered a team of seven young residents to document blighted properties in the neighborhood and how blight affects the community residents. After documenting over 250 homes in two days, the team presented their findings to Detroit City Hall and the Michigan State Legislature. The experience gave the kids an insight into the impact of blight on their neighborhood and enabled them to act as community leaders.*

MOTOR
CITY
MAPPING

## "BLEXTING" - A COMBINATION OF BLIGHT AND TEXTING

# DATA COLLECTION

In November 2013, the Blight Removal Task Force, in partnership with Michigan Nonprofit Association, Data Driven Detroit, and Loveland Technologies, began a physical survey to gather property condition data for all 380,000 parcels in the city. The goal of the Motor City Mapping (MCM) project was to create a comprehensive database of detailed information including the condition of each and every property in Detroit. The process is designed to be scalable, repeatable, and real-time to ensure that relevant data will be available for policy makers both now and in the future.

Data was collected via a mobile application in a process nicknamed "Blexting," (a combination of *blight* and *texting*) that built on prior data from the 2009 Detroit Residential Parcel Survey (DRPS). At that time, the 2009 survey was the first of its kind, conducted by the public-private Detroit Data Collaborative, a team effort of the Detroit Office of Foreclosure Prevention and Response, Michigan Community Resources, University of Michigan Ginsburg Center, and Data Driven Detroit. The 2014 MCM survey expanded on the 2009 survey to include non-residential properties.

# DETROIT RESIDENTS DID THE WORK

Over a period of ten weeks, a team of approximately 150 resident surveyors and volunteer drivers were assigned to quarter-mile areas (nicknamed "microhoods") to document property conditions. The surveyors photographed the front of each property and answered a series of specific questions related to the its condition. The survey feedback included observations on occupancy, lot vacancy, fire damage, presence of dumping and the nature of the use of the property (residential, commercial, etc.). Once the surveyor completed the series of questions, the information was instantaneously uploaded via a live-stream feed to 'mission control', where associates performed a quality control check on all data submitted from the field. *The full questionnaire is located in the Appendix section of this report.*

## MOTOR CITY MAPPING "BLEXTING" FIELD SURVEY

**IS THERE A STRUCTURE ON THE SITE?**

**IF YES,**

**PLEASE ANSWER THE FOLLOWING QUESTIONS FOR STRUCTURES:**

- **What is this site used for?**
- **Number of Units (Only for Residential and Mixed Use Residential/ Commercial)**
- **What is the condition of the structure?**
- **Is the structure occupied?**
- **Is the structure open and in need of boarding?**
- **Is the structure fire damaged?**
- **Is there dumping on the site?**

**PLEASE RATE THE FOLLOWING ELEMENTS TO DETERMINE THE STRUCTURAL ASSESSMENT RATING:**

- **Building Walls and Foundation**
- **Roof**
- **Doors and Windows**
- **Siding/Masonry**
- **Chimney and Gutters**

**IF NO,**

**PLEASE ANSWER THE FOLLOWING QUESTIONS FOR LOTS:**

- **What is this site used for?**
- **Is the lot improved or unimproved?**
- **Is the lot maintained?**
- **Is there dumping on the site?**

# MOTOR CITY MAPPING FIELD SURVEY PROCESS

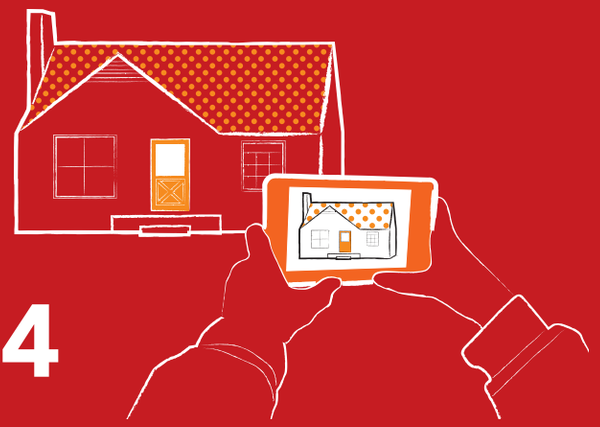
**Loveland Technologies and Data Driven Detroit joined various representatives of the Blight Task Force to plan the massive citywide survey effort.**

**1** A group of urban planners, technology experts, and residents created the structure and plan for the Motor City Mapping effort.

**2** With the Blexting system in-hand, surveyors were trained to use the application.

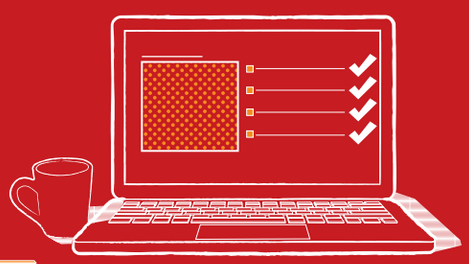
**3** Motor City Mapping organizers choreographed the process to survey to every single property in Detroit.

**4** After the mapping process was defined, teams of two gathered information on each and every property in the city.



**5** Soon there was a small army of survey teams working across the city, submitting data in real-time.

**6** As data came in from the field, quality control associates checked data for accuracy and completeness.



**7** Information from 24 separate databases (ex. tax information, utilities, postal, etc) were then added to the Motor City Mapping database, enhancing the information for nearly every property in the city.

## BLIGHT IS…

…a public nuisance.

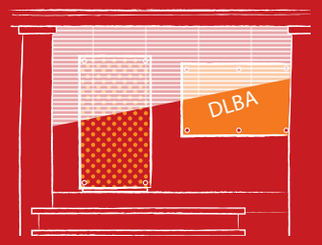
…an attractive nuisance.

…fire damaged or is otherwise dangerous.

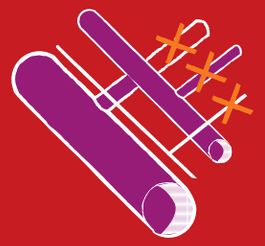
…already on Detroit's BSEED Demolition list.

…has code violations posing a severe and immediate health or safety threat.

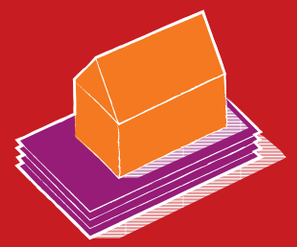
…open to the elements and trespassing.




…owned or is under the control of a land bank.



…has had the utilities, plumbing, heating or sewerage disconnected, destroyed, removed, or rendered ineffective.



…a tax-reverted property.

…has been vacant for 5 consecutive years, and not maintained to code.



# [ DEFINING BLIGHT ]

There are several technical and legal definitions of blight and then there is the daily experience of blight. One trait that is common across all definitions and experiences of blight is that it is caused by neglect. A small problem becomes bigger over time. Without intervention, sooner or later, blight spreads and takes over a block, then a neighborhood, then a district, and then a city. The Task Force's definition and methodology for classifying property as "Blight" incorporates the concepts of physical blight, economic blight, the public's interest in protecting the health, safety, and general welfare of people in its communities, and the preservation of property values.

Michigan law defines a "blighted property" as one that meets any of the following conditions as determined by an applicable "governing body":

- A public nuisance
- An attractive nuisance *(see below)*
- A fire hazard or is otherwise dangerous
- Has had utilities, plumbing, heating or sewerage disconnected, destroyed, removed, or rendered ineffective
- A tax-reverted property
- Owned or under the control of a land bank
- Has been vacant for 5 consecutive years and not maintained to code
- Has code violations posing a severe and immediate health or safety threat

***Definition of an Attractive Nuisance**: A landowner may be held liable for injuries to children trespassing on the land if the injury is caused by a hazardous object or condition on the land that is likely to attract children who are unable to appreciate the risk posed by the object or condition. (i.e. abandoned cars, poles of lumber, debris, etc.)*

# TASK FORCE DEFINITION OF "BLIGHT"

Using the State of Michigan's definition of "blighted property" as a starting point, the Task Force added elements from the Detroit Ordinance governing "dangerous buildings".

The Task Force definition includes properties that are:

- Open to the elements and trespassing, and
- On Detroit's Buildings, Safety Engineering, and Environmental Department (BSEED) Demolition list.

Accordingly, properties that are exposed to the elements, are not structurally sound, are in need of major repairs, are fire damaged, or have essentially been turned into a neighborhood dumping ground, were classified as "blight" by the Task Force.

While the Task Force recognized that there are other aspects that border on any reasonable definition of blight such as graffiti, un-manicured boulevard grass and litter, the scope of this report addresses **blight** based on the Task Force definition.

# INDICATORS OF FUTURE BLIGHT

In addition to identifying structures and lots that met the Task Force definition of "Blight," we identified other properties which possessed indicators of becoming future blight ("Blight Indicators"). Properties with "Blight Indicators" are those properties and structures which did not meet our definition of "Blight," yet, had the following characteristics:

- Were unoccupied and/or abandoned (regardless of duration), or
- Were publicly owned by local or state authorities, or owned by Government Sponsored Entities (such as Fannie Mae and Freddie Mae),

While a small number of these properties may appear in fair condition today, there is a high probability that they will become blighted in the near future and need to be removed. The Task Force recommends the approach of further inspection, data gathering and analysis to determine the appropriate intervention.

## BLIGHTED VACANT LOTS

The Motor City Mapping field survey identified vacant lots showing evidence of significant dumping that need immediate attention. These parcels are considered an attractive nuisance and therefore meet the Task Force definition of blight.

The Task Force compiled 24 data sets from a variety of public and private sources. The Motor City Mapping team created a comprehensive database merging the field survey observations with 16 of the 24 data sets. The remaining eight data sets were not used to define the Task Force's definition of blight but will be used in the next phase of the MCM project. For the first time in Detroit's history, a deep and complete picture of each property in the city is now accessible for strategic, tactical analytics and decision making.

**Data Sets Used To Define Blight (16):**

- 2009 Detroit Residential Parcel Survey
- Wayne County Register of Deeds - Sheriff's Sales, Land Contracts, Mortgages, Warranty Deeds, and Quit-Claim Deeds, 2012 - 2013
- Wayne County Treasurer - Properties with Reverter Clause/Sold At Tax Auction, 2011 - 2013
- City of Detroit Buildings, Safety Engineering, and Environmental Department (BSEED) Dangerous Buildings List
- City of Detroit Fire Escrow Fund Properties
- Valassis VNEF Plus Address Database (Confirmed by USPS, January, 2014)
- Fannie Mae REO Inventory
- Freddie Mac REO Inventory
- Detroit Parcel Inventory
- 2014 Historic Resource Survey
- D3 Historic Districts and Eligible Districts Dataset
- 2013 City of Detroit Assessor's File
- Michigan Land Bank Fast Track Authority - Current Inventory
- Detroit Land Bank Authority - Current Inventory
- Detroit Future City Current Framework Zones
- 2014 Tax-Delinquent Properties

**Data Sets Collected But Not Used To Define Blight (8):**

- DTE Energy Gas and Electric Shut-offs
- DWSD Water Shut-offs
- Clear Corps Intervention Locations
- Wayne County Treasurer - Properties Reaching Tax Auction, 2002 - 2013
- Detroit Fire Department Incidents, 2012 - 2013
- SEMCOG Building Footprints
- Detroit Future City 10- and 50-year Future Land Use Framework
- Targeted Investment Areas

**ONE TRAIT THAT IS COMMON ACROSS ALL DEFINITIONS AND EXPERIENCES OF BLIGHT IS THAT BLIGHT IS CAUSED BY NEGLECT.**

This table demonstrates how the Task Force's definition of blight was mapped to the data sets and field survey information captured through Motor City Mapping.

| BLIGHT DEFINITION CATEGORY | DATA SETS (MOTOR CITY MAPPING) |
|---|---|
| Pubic Nuisance | Conditions (Suggest Demo/poor) |
| Attractive Nuisance | Dumping |
| Fire Hazard | Fire Damage |
| Utilities Disconnected | DTE Data |
| Tax Reverter | Public Datasets |
| Land Bank Controlled | Public Datasets |
| Vacant | Basket Index (Survey, DTE, Postal Data) |
| Code Violations | Administrative Dataset |
| Open to the Elements | Survey Dataset |
| BSEED Demo List | BSEED Dataset |

## MOTOR CITY MAPPING DATA RESULTS

| PARCEL TYPE | BLIGHT PUBLIC | BLIGHT PRIVATE | BLIGHT TOTAL | NON-BLIGHT PUBLIC | NON-BLIGHT PRIVATE | NON-BLIGHT TOTAL |
|---|---|---|---|---|---|---|
| STRUCTURE RESIDENTIAL | 24,064 | 48,971 | 73,035 | 1,602 | 170,970 | 172,572 |
| | 33% | 67% | 100% | 1% | 99% | 100% |
| STRUCTURE COMMERCIAL | 913 | 4,558 | 5,471 | 339 | 12,152 | 12,491 |
| | 17% | 83% | 100% | 3% | 97% | 100% |
| TOTAL WITH STRUCTURE | 24,977 | 53,529 | 78,506 | 1,941 | 183,122 | 185,063 |
| | 32% | 68% | 100% | 1% | 99% | 100% |
| VACANT LOT | 3,680 | 2,455 | 6,135 | 53,817 | 54,081 | 107,898 |
| | 60% | 40% | 100% | 50% | 50% | 100% |
| TOTAL | 28,657 | 55,984 | **84,641** | 55,758 | 237,203 | 292,961 |
| | 34% | 66% | 100% | 19% | 81% | 100% |

| PARCEL TYPE | TOTAL PUBLIC | TOTAL PRIVATE | TOTAL | % BLIGHTED |
|---|---|---|---|---|
| STRUCTURE RESIDENTIAL | 25,666 | 219,941 | 245,607 | 30% |
| | 10% | 90% | 100% | |
| STRUCTURE COMMERCIAL | 1,252 | 16,710 | 17,962 | 30% |
| | 6.97% | 93% | 100% | |
| TOTAL WITH STRUCTURE | 26,918 | 236,651 | 263,569 | 30% |
| | 10% | 90% | 100% | |
| VACANT LOT | 57,497 | 56,536 | 114,033 | 5% |
| | 50% | 50% | 100% | |
| TOTAL | 84,415 | 293,187 | **377,602** | 22% |
| | 22% | 78% | 100% | |

Motor City Mapping (MCM) successfully surveyed 378,136 of the 380,233 properties within the 139 square miles of the city. The difference of 2,097 properties represents those that could not be viewed from the street. Data Driven Detroit confirmed that data received from **377,602** surveys accurately matched third party data sets. The table above highlights several interesting facts including **84,641** structures and vacant lots need intervention.

**1 IN 2**

VACANT LOTS ARE OWNED BY PUBLIC AGENCIES

**30% LOTS**

(114,000) OF THE CITY'S PROPERTIES SURVEYED WERE LOTS

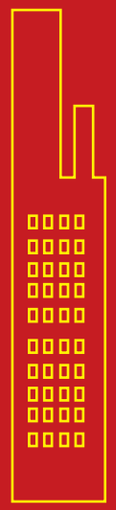
# SURVEYED PROPERTIES

377,602 PROPERTIES SURVEYED WITHIN THE CITY

# 377K

**70% WITH STRUCTURES**

(264,000) OF THE CITY'S PROPERTIES SURVEYED INCLUDES A STRUCTURE



**1 IN 3**

RESIDENTIAL + COMMERCIAL STRUCTURES ARE BLIGHTED

**94%**

OF PUBLICLY OWNED RESIDENTIAL STRUCTURES ARE BLIGHTED



# [ WHAT WE FOUND ]

Of the 84,641 structures and vacant lots in need of intervention:

**40,077 structures clearly meet the Task Force definition of "blight" and are recommended for immediate removal.** Community engagement will be required to confirm removal or otherwise address significant remedial action for these properties.

**38,429 structures have "Blight Indicators" and need further evaluation.** The Strategic Assessment Triage Tool, discussed in Chapter 5, can be used to guide the evaluation process. The spectrum of interventions includes rehabilitation, removal, or securing.

**At least 6,135 vacant lots showed evidence of dumping and need immediate attention.** (Given the record breaking snowfall of 2014, the surveyors had difficulty getting an accurate identification of the vacant lot conditions.)

**40,077**

STRUCTURES THAT MEET TASK FORCE DEFINITION OF BLIGHT

**+ 38,429**

STRUCTURES WITH INDICATORS OF FUTURE BLIGHT

**78,506**

STRUCTURES WITH INDICATORS OF FUTURE BLIGHT

**+ 6,135**

BLIGHTED VACANT LOTS

**84,641**

TOTAL PROPERTIES MEETING THE BLIGHT TASK FORCE DEFINITION OF BLIGHT OR HAVE BLIGHT INDICATORS (INTERVENTIONS NEEDED)

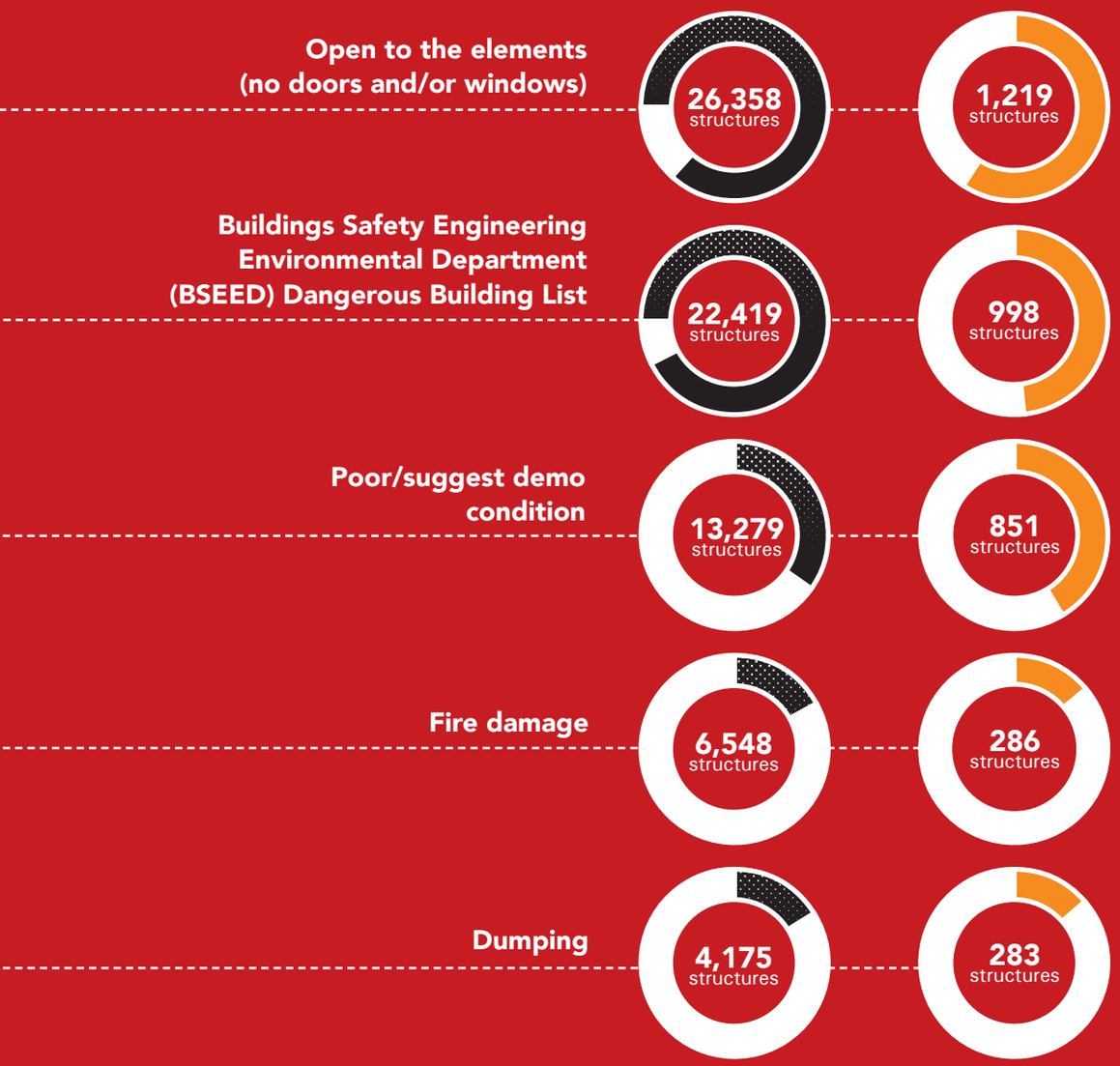
# 40K

40,077 structures are recommended for removal as they meet the Task Force definition of blight (Residential and non-residential structures)

**RESIDENTIAL STRUCTURES (38,021)**

**NON-RESIDENTIAL STRUCTURES (2,056)**

## DATA SET CATEGORIES



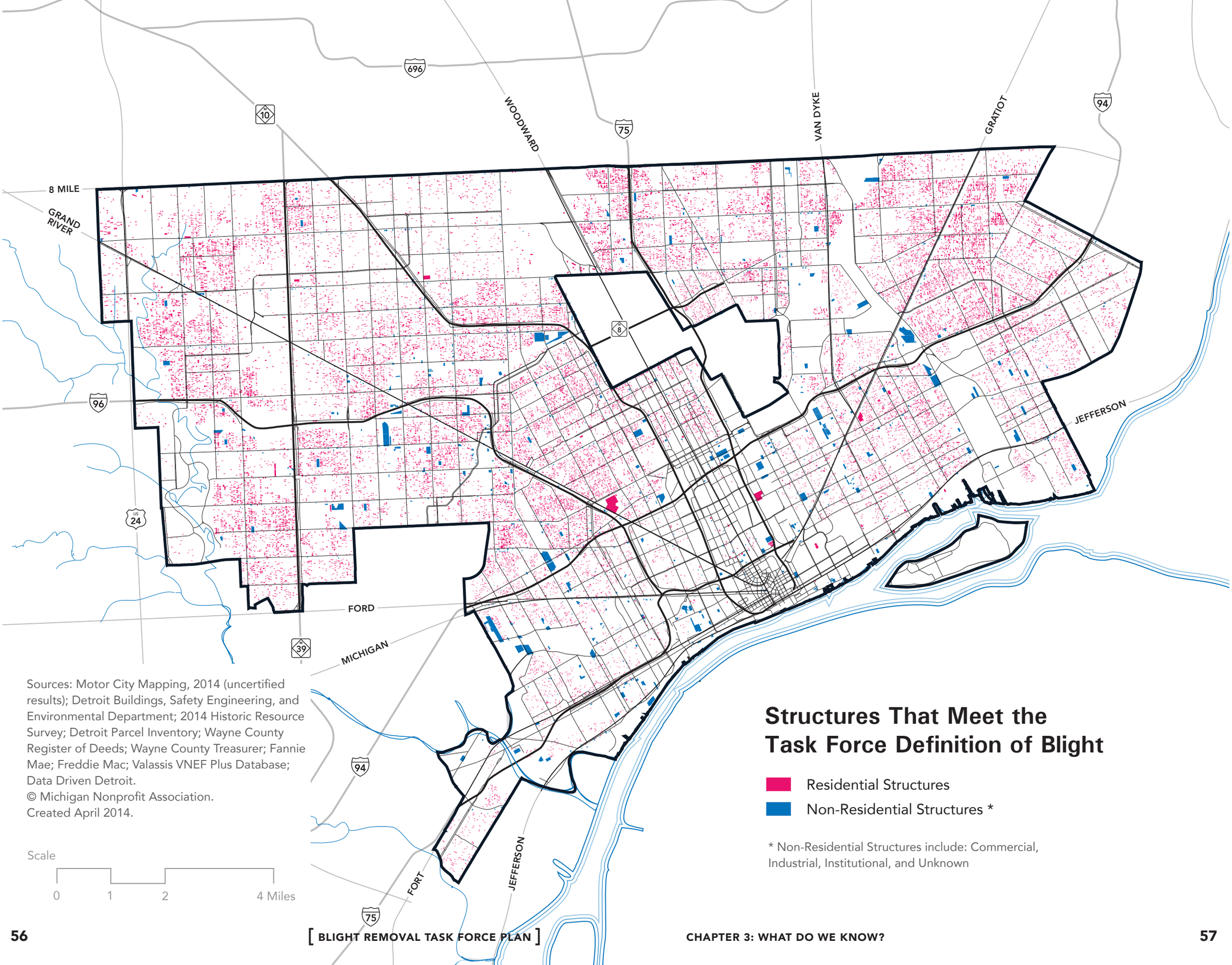

NOTE: Structures fall into one of these categories but may fall into more than one. For example, a structure may be in poor/suggested demo conditions and have fire damage.

## 40,077 STRUCTURES MEET TASK FORCE DEFINITION OF BLIGHT

The MCM team identified 40,077 structures that meet the Task Force definition of blight and are candidates for removal. Prior to any building being brought down, it is essential to engage with residents and the community to confirm removal is the appropriate intervention. *(Specific outlines for community engagement can be found in this report in Chapter 2.)*

Structures that are candidates for removal meet any one of the following criteria:

- Reported with Poor or Suggest Demo external condition in the MCM Survey
- Fire Damaged
- Listed in the Buildings, Safety Engineering, and Environmental Department Dangerous Buildings inventory
- Open to the elements or trespass
- Significant dumping




Sources: Motor City Mapping, 2014 (uncertified results); Detroit Buildings, Safety Engineering, and Environmental Department; 2014 Historic Resource Survey; Detroit Parcel Inventory; Wayne County Register of Deeds; Wayne County Treasurer; Fannie Mae; Freddie Mac; Valassis VNEF Plus Database; Data Driven Detroit.
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Scale
0 1 2 4 Miles

## Structures That Meet the Task Force Definition of Blight

- Residential Structures
- Non-Residential Structures *

* Non-Residential Structures include: Commercial, Industrial, Institutional, and Unknown

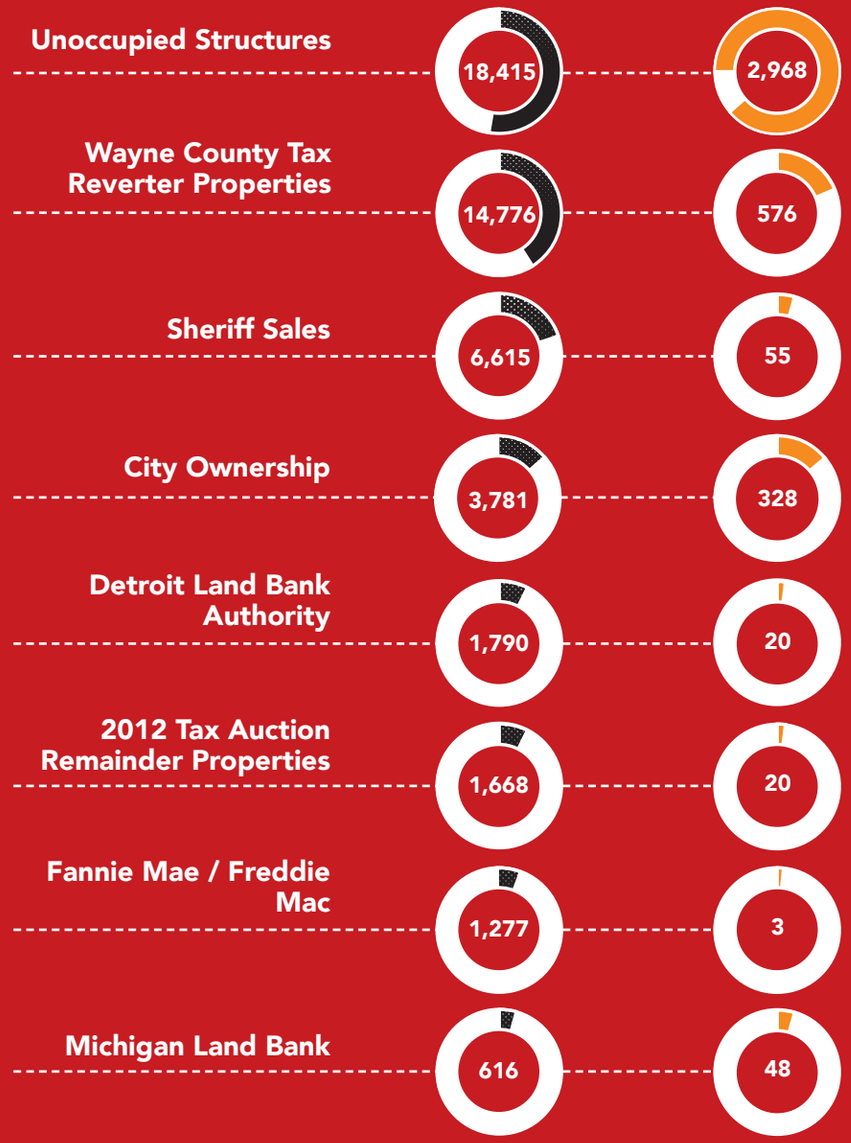
# 38K

38,429 structures with blight indicators and identified for further analysis (Residential and non-residential structures)

RESIDENTIAL STRUCTURES (35,014)

NON-RESIDENTIAL STRUCTURES (3,415)

## DATA SET CATEGORIES




NOTE: Structures fall into one of these categories but may fall into more than one. For example, a structure may be both unoccupied and a Michigan Land Bank property.

## 38,429 STRUCTURES WITH INDICATORS OF FUTURE BLIGHT

Properties in this category meet criteria that are referred to as properties having Blight Indicators and will need  further evaluation to identify the appropriate intervention. Structures within this category meet any one of the following criteria:

- Unoccupied and/or abandoned
- Land Bank & city ownership
- Wayne County ownership (including reverter clause)
- Sheriff's Deed
- Fannie Mae or Freddie Mac ownership

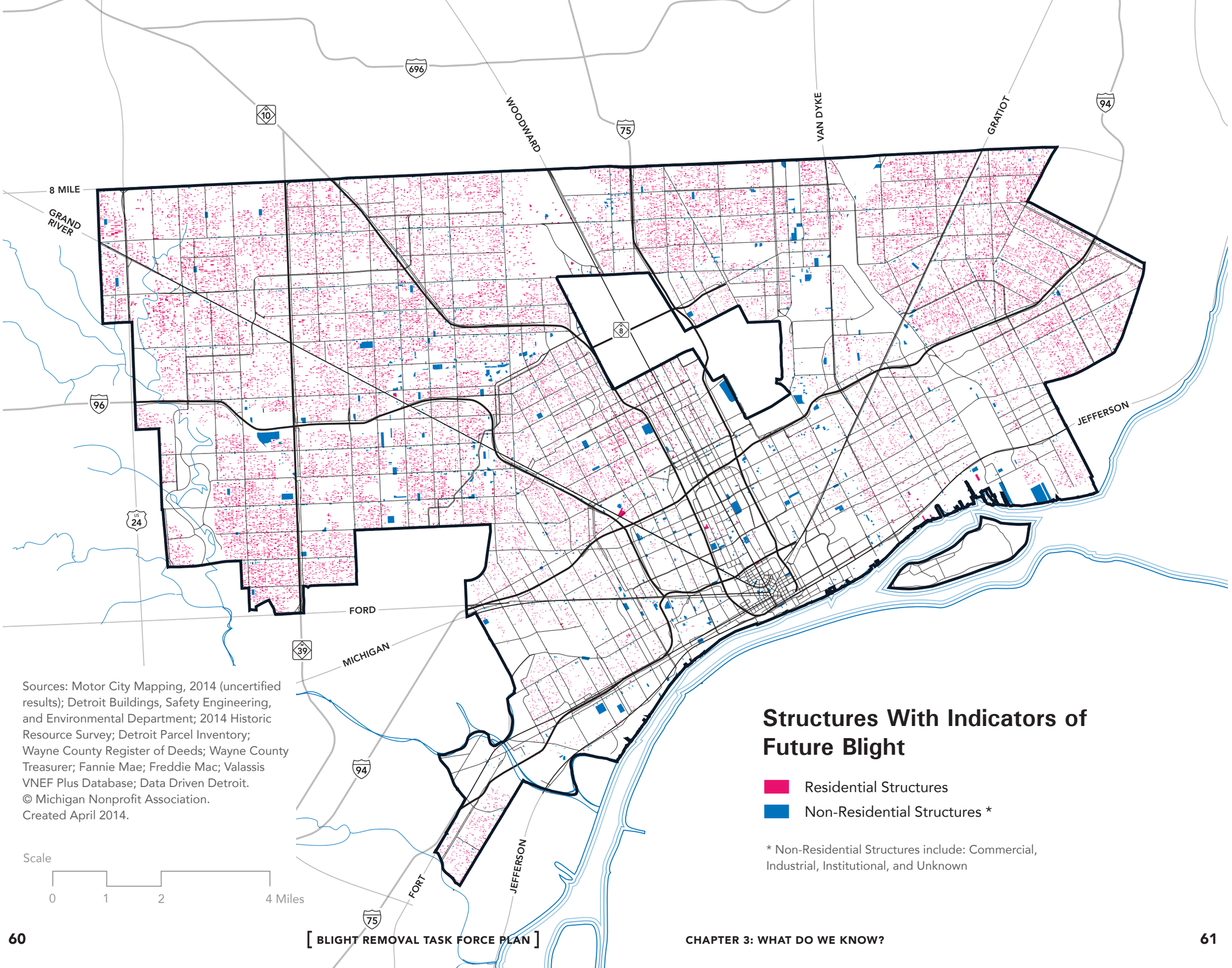
While these properties may otherwise be considered non-blight (i.e., arguably in good or fair condition) today, based on these characteristics, there is a high probability that they will devolve into Blight in the near future and need to be removed. The Task Force recommends further inspection and data gathering to confirm this intervention approach.

While MCM has collected a robust database of property information, additional factors need to be considered prior to deciding how to intervene on a these specific properties. In addition to getting direct feedback from the community, the Task Force recommends using the Strategic Assessment Triage Tool (SATT) to gather additional information. *(The SATT tool is explained in more detail in Chapter 5.)*

Properties included in the group of properties with "Blight Indicators" and in need further analysis does not mean that doing nothing is an option. For all of these structures, one of three interventions will be needed: removal, through demolition (utilizing deconstruction in some cases), restoration back to code, or, secure and maintain until a new use can be found.

During conversations with local deconstruction and demolition contractors, the Task Force learned that structures in this category have a high likelihood of resulting in candidates for removal. Primarily, contractors find that rehabilitation is cost prohibitive from a market perspective.  While there are circumstances where individuals, government and developers  may pay or invest more than the market value to restore a structure, such instances are rare. One example would be a home inherited from a family member which has sentimental value. Another example is when developers are trying to bolster the real estate market in a particular neighborhood. Based on the information gathered, the Task Force assumed that 80-90 percent of properties with Blight Indicators (or approximately 32,000 of the structures in this category) will need to be removed in the near future.

This would result in an anticipated blight removal candidate pool of approximately 72,000 structures (40,077 parcels that meet the definition of blight + 32,000 parcels that represent 80-90 percent of the parcels with blight indicators).




Sources: Motor City Mapping, 2014 (uncertified results); Detroit Buildings, Safety Engineering, and Environmental Department; 2014 Historic Resource Survey; Detroit Parcel Inventory; Wayne County Register of Deeds; Wayne County Treasurer; Fannie Mae; Freddie Mac; Valassis VNEF Plus Database; Data Driven Detroit.
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Created April 2014.

Scale

0 1 2 4 Miles

## Structures With Indicators of Future Blight

- Residential Structures
- Non-Residential Structures *

* Non-Residential Structures include: Commercial, Industrial, Institutional, and Unknown

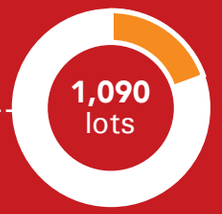
# 6,135

**TOTAL NUMBER OF VACANT LOTS WITH DUMPING**

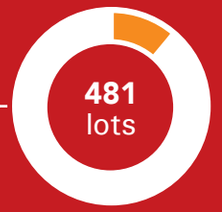
**VACANT LOTS**

## DATA SET CATEGORIES

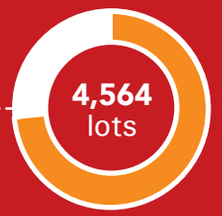
**Private use**
**(188 acres total)**




**Public Use**
**(94 acres total)**



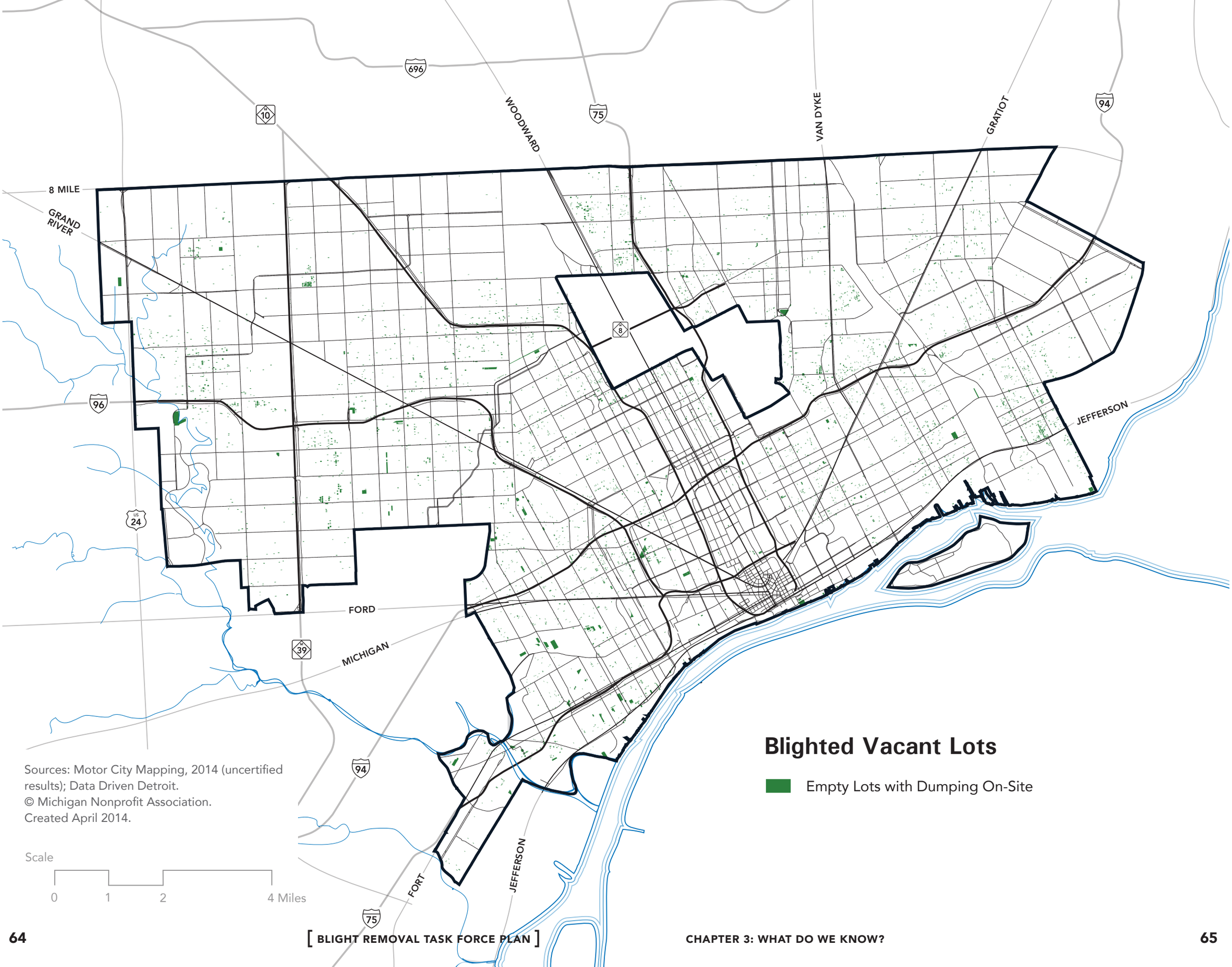

**Unknown Use**
**(508 acres total)**



## 6,135 BLIGHTED VACANT LOTS

Thousands of overgrown, vacant lots must be factored into Detroit's blight elimination strategy. Often littered by years of illegal dumping, these lots also present numerous health, public safety, and economic challenges that must be addressed alongside the 78,506 blighted structures. Since the MCM team gathered its information in the heart of a Michigan winter with record-setting snowfalls, most vacant parcels were covered in several inches of snow. The surveyors had difficulty recording an accurate indication of the site conditions. Nonetheless, Data Driven Detroit is confident that at least 6,135 vacant lots showed evidence of dumping that need immediate attention. Further input from the community over the spring and summer will refine the number of vacant lots with blight indicators.

Blight mitigation must include residential structures, commercial structures, industrial structures, and vacant lots. The Blight Removal Task Force believes that all four of these components are crucial and must be worked on together to create the greatest amount of change as we work toward the goal of a leaner, cleaner, and healthier Detroit.


696
10
WOODWARD
75
VAN DYKE
GRATIOT
94
8 MILE
GRAND RIVER
8
96
JEFFERSON
24
FORD
39
MICHIGAN
94
FORT
JEFFERSON
75
Blighted Vacant Lots
Empty Lots with Dumping On-Site
Sources: Motor City Mapping, 2014 (uncertified results); Data Driven Detroit.
© Michigan Nonprofit Association.
Created April 2014.
Scale
0
1
2
4 Miles

## SURVEYED PROPERTIES

**377,602 PROPERTIES SURVEYED WITHIN THE CITY**

The Motor City Mapping successfully surveyed 378,136 of the 380,233 parcels within the 139 square miles of the city. The difference of 2,097 parcels represents those properties that could not be viewed from the street. Data Driven Detroit confirmed that data received from 377,602 surveys accurately matched third party data sets.

**STRUCTURES**

263,569 STRUCTURES

**LOTS**

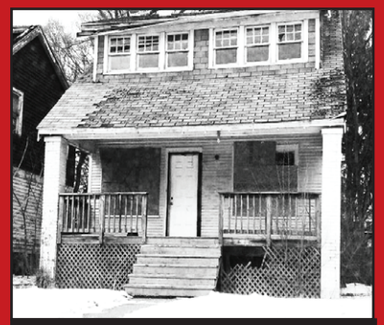
114,033 LOTS

## BLIGHT = 84,641 PARCELS

**78,506 STRUCTURES & 6,135 LOTS REQUIRE AN INTERVENTION**

**STRUCTURES THAT MEET TASK FORCE DEFINITION OF BLIGHT**

40,077 STRUCTURES



**STRUCTURES WITH INDICATORS OF FUTURE BLIGHT**

38,429 STRUCTURES

**BLIGHTED VACANT LOTS**

6,135 LOTS



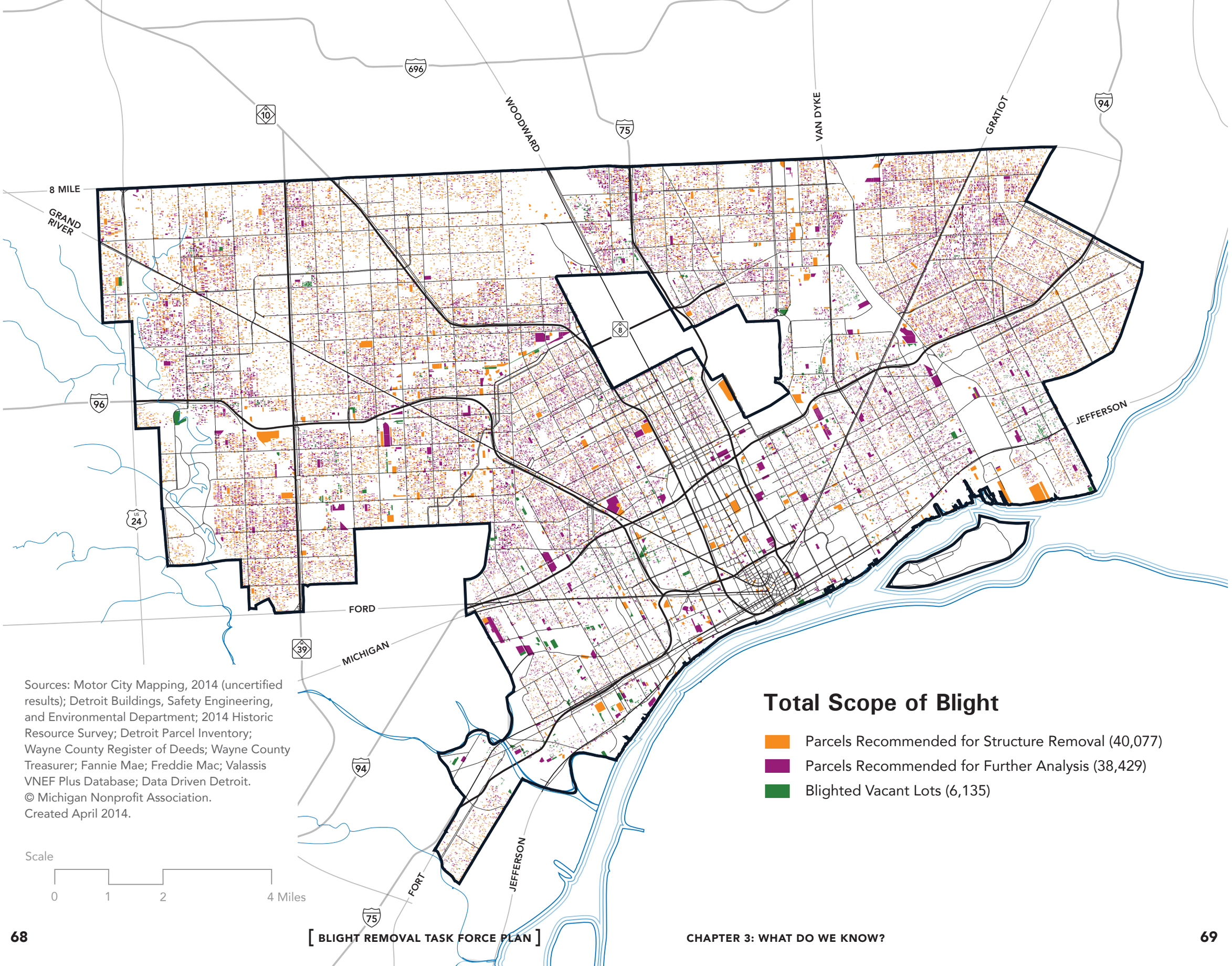
# THE BLIGHT SPECTRUM: TOTAL SCOPE OF BLIGHT

**There are a total of 84,641 blighted structures and vacant lots in the City of Detroit.**

**Based on the collected data, 40,077 structures clearly meet the Task Force definition of "blight" and are recommended for immediate removal.** Community engagement will be required to confirm removal or otherwise address significant remedial action for these properties. The Detroit Land Bank Authority has already started this validation process working directly within the Hardest Hit Fund neighborhoods.

**An additional 38,429 structures with "Blight Indicators" will need further evaluation.** The Strategic Assessment Triage Tool, discussed in Chapter 5, can be used to guide the evaluation process. The spectrum of interventions includes rehabilitation, removal, or securing.

Given the record breaking snowfall of 2014, the surveyors had difficulty getting an accurate identification of the vacant lot conditions. **None the less, the data team is confident that at least 6,135 vacant lots showed evidence of dumping and need immediate attention.**




Sources: Motor City Mapping, 2014 (uncertified results); Detroit Buildings, Safety Engineering, and Environmental Department; 2014 Historic Resource Survey; Detroit Parcel Inventory; Wayne County Register of Deeds; Wayne County Treasurer; Fannie Mae; Freddie Mac; Valassis VNEF Plus Database; Data Driven Detroit.
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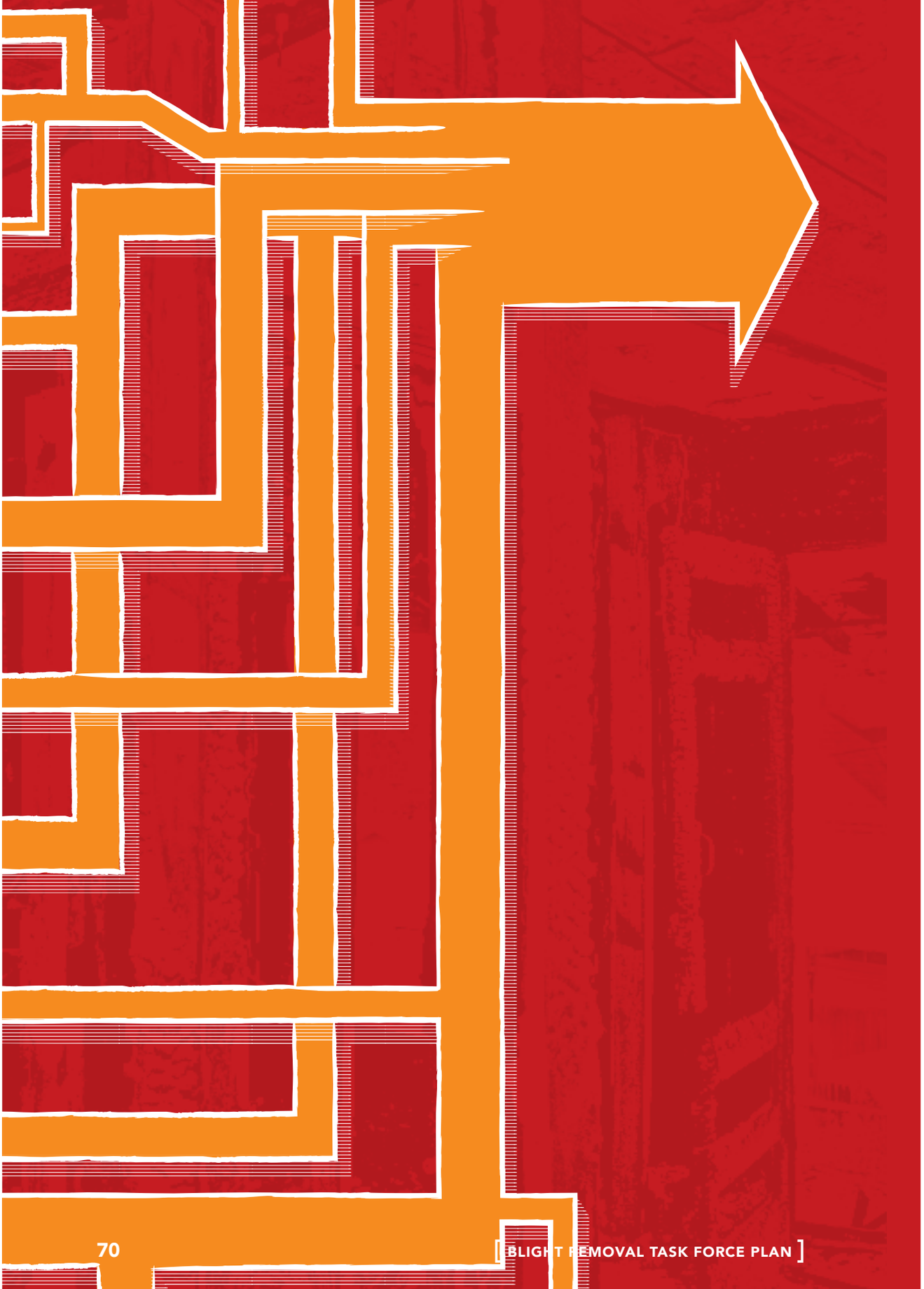
Scale

0 1 2 4 Miles

## Total Scope of Blight

- Parcels Recommended for Structure Removal (40,077)
- Parcels Recommended for Further Analysis (38,429)
- Blighted Vacant Lots (6,135)



# [ IMPROVING OUR KNOWLEDGE ]

More than 14 agencies interact with properties, safety, and public works in Detroit. These groups create rich databases through their daily activities, such as issuing permits for construction or fixing a streetlight. In aggregate, these datasets can paint a detailed picture of what is going on in the city from a host of perspectives. However, each of these organizations has its own practices and challenges in collecting and updating information, meaning that databases are often compartmentalized or locked into the department in which they were created. This situation is not unique to Detroit, but because of the scale of the challenges facing our city there is a pressing need to change the current systems.

The Mayor's decision to bring many of these agencies together under the Department of Neighborhoods is an important first step toward better coordination. Sharing data effectively is the next step. The Mayor has appointed the first-ever Cabinet level Chief Information Officer (CIO) to lead this effort.

## WHAT IS THE PROBLEM?

Detroit is suffering from an "information crisis" that has profound impacts on residents. Each city agency has its own proprietary dataset containing information with varying degrees of accuracy. Information is collected and maintained in different ways across different agencies for different purposes, resulting in disconnected datasets. For example, the Detroit Fire Department cannot easily see BSEED's Dangerous Buildings list, meaning fire fighters often rush into buildings not knowing the structure is condemned. In addition, this information is typically not shared with the general public.

Another complicating factor is that too much information is still collected and stored in paper form. This not only hampers the work of city agencies, but also prevents residents and community groups from providing and receiving crucial information. Manually matching one set of data to another wastes time, resources, and expertise at a moment when funding and specialized personnel have been in short supply. This lack of visibility severely limits the city and its departments ability to prioritize actions and solutions, as well as share valuable information with the general public.

THE PUBLIC DASHBOARD IS AN INTERACTIVE, WEB-BASED DATA EXPLORATION TOOL THAT IS OPEN TO THE PUBLIC.

## WHAT ARE THE SOLUTIONS?

The answer to Detroit's information crisis is to build a system that facilitates communication and the exchange of information among government departments and residents. The Mayor's CIO will lead this effort, which includes implementing Phases II and III of Motor City Mapping.

The Motor City Mapping (MCM) survey (Phase I) was the first step in understanding the current conditions of blight in Detroit. The second phase of MCM will allow residents to provide critical information as well as analytical tools to the Detroit Land Bank Authority. This expanded capability will allow city departments and residents to gain visibility and understanding of the city with a depth and breadth never experienced in any municipality.

The Task Force recommends the City and DLBA implement Motor City Mapping Phases II and III. Phase II of Motor City Mapping is the creation of a centralized, public database. This database will contain information from the Motor City Mapping Phase I, plus data from a wide variety of sources, including numerous municipal departments. The project is planned for launch in late spring 2014 and scheduled to be completed in fall 2014.

Phase II has four deliverables:

- **Public Dashboard**
- **Mobile App**
- **Land Bank Tool**
- **Data Management Team**

Phase III delivers an expanded site control dashboard and connects all municipal departments whose functions and responsibilities are associated with real property.

**THE PUBLIC DATABASE SHOULD BE AS TRANSPARENT AS POSSIBLE, WHILE RESPECTING THE PRIVACY CONCERNS ASSOCIATED WITH THE DISPLAY OF SENSITIVE DATA.**

**PUBLIC DASHBOARD.** The public dashboard will be an interactive, web-based data exploration tool that is open to the public. It will give residents, academics, community groups, philanthropic organizations, and businesses a deeper understanding of the city than has ever been possible before.

The centerpiece of the dashboard will be an interactive map of the City of Detroit, which will be familiar to users of Google or Bing Maps. Detroit-based Loveland Technologies will use its maps from the website, whydontweownthis.com, as the foundation for Motor City Mapping. Layered on top of this primary map will be parcel-shaped files displaying the boundaries of each property in the entire city. These parcel-shaped files are the building blocks of most public data.

Users of the dashboard will be able to explore the physical aspects of the city through a multitude of publicly available datasets that focus on structure age, recent sale price, current assessments, and ownership information. While all of this information is currently housed within various public records, there has never before been an all-encompassing, intuitive, and easy-to-use platform to access them.

The public dashboard will also serve as a real-time forum for discussion related to property characteristics, history, and conditions. Each property will display a current photo, (similar to Google Street View) along with property data sourced from the Motor City Mapping property survey. The system will be designed to allow residents to update pictures and property information by submitting their feedback on-line. This feedback will be vetted for accuracy and relevancy before being posted to the database. The public dashboard will serve as a critical link between the residents and the City of Detroit.

In addition, the dashboard will serve as a communication platform between city departments as well as related governmental agencies. City departments will post updated information demonstrating recent activities such as location and schedule of demolition activity and upcoming projects.

The Task Force recommends that the Detroit Land Bank Authority and Loveland Technologies work with District Managers to provide training of the public dashboard at community meetings.

**MOBILE APP.** A mobile version of the public dashboard is planned for release by fall 2014. The mobile app, named "Blexting" (short for blight + texting), will also allow users to submit photos and property condition information in real-time from the field. This will permit thousands of people to contribute to a shared, visible, and accessible database of individual property characteristics and neighborhood conditions. This open and shared platform will house rich and valuable information allowing for better decision making and prioritization that is light years ahead of today's crippled process.

**LAND BANK TOOL.** Phase II will also include a specific Toolkit within the dashboard allowing the Detroit Land Bank Authority (DLBA) to analyze the city's data in new ways. This technology will enable the DLBA to dramatically improve the management of large-scale projects and its growing inventory of property.

For example, in March 2014 the DLBA began the process of identifying homes for demolition through the federal Hardest Hit Funds program. The Land Bank has been using the data collected from the Motor City Mapping survey and the community to help it make decisions about which structures should be removed. The Toolkit will allow the DLBA to quickly explore and analyze large amounts of data to understand a new range of variables such as utility information and property occupancy.

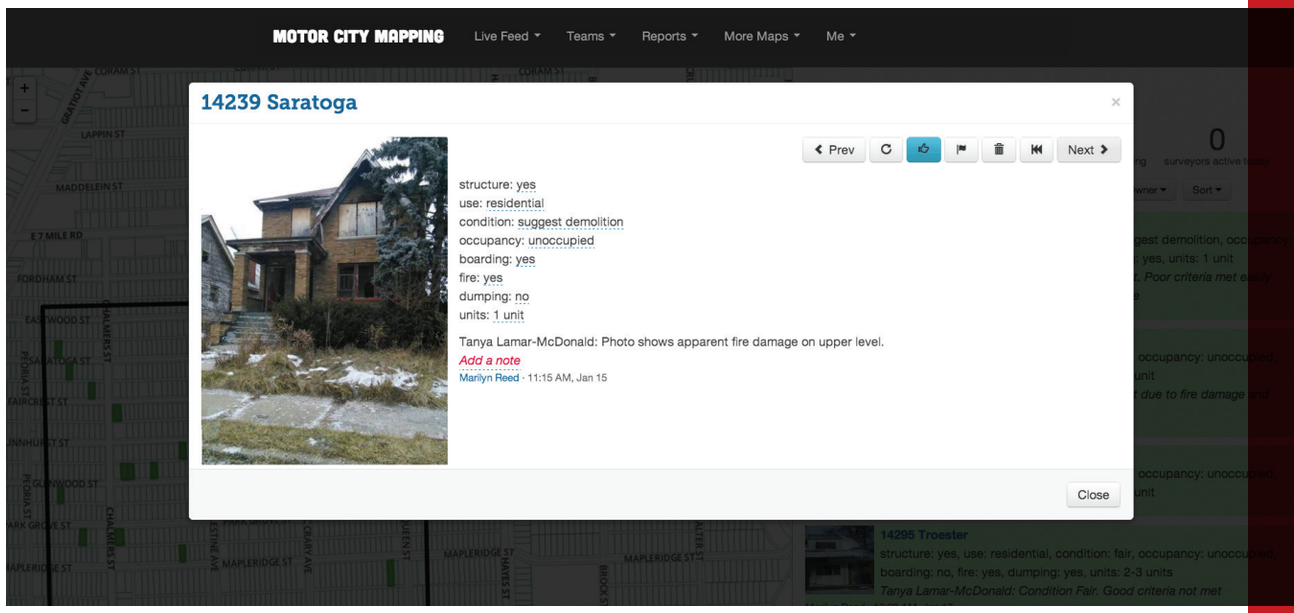
The Toolkit will feature technology to assist the DLBA to manage large scale demolition operations. Real-time information such as property demolition selection, contractor assignment, and project status will be invaluable. In addition, the toolkit will allow the DLBA to quickly and easily share and move data among numerous partners in a variety of formats. This information can be generated with a click of a button.

**DATA MANAGEMENT TEAM.** Data Driven Detroit (DDD), a local subsidiary of Michigan Nonprofit Association, is dedicated to providing accurate, high quality information to help shape data-driven decision making. DDD will fill the role of Data Manager. DDD's expertise in data management, their deep involvement with a number of community organizations, and their commitment to transparency make them a critical partner in this endeavor.

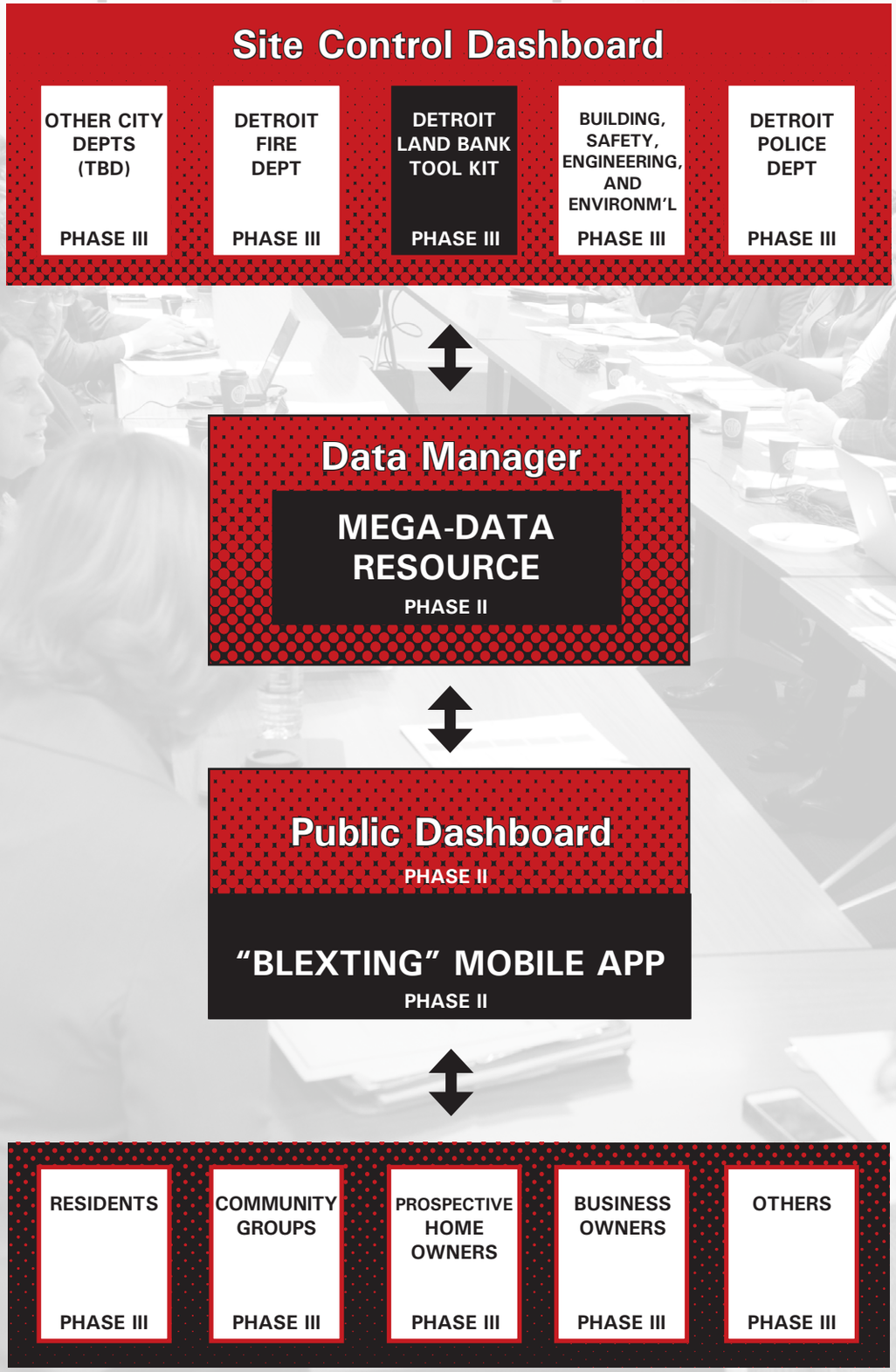
A database is only useful when it houses accurate, up to date, complete content. DDD will take on two principal responsibilities; maintaining the integrity of the database, and teaching best practices and data literacy to all those involved, including members of the public.

The data management team will monitor and evaluate the information provided by members of the public through the mobile app and public dashboard. They will also work closely with city departments to help improve data collection processes, increasing efficiency and further enrich the MCM database.

Through community meetings, the team will work with local residents and community groups to promote the collection of information by providing demonstrations and public dashboard and mobile app training.



**PHASE III MOTOR CITY MAPPING.** The Task Force recommends Detroit expand the MCM tool beyond the DLBA to all municipal departments that deal with property. This will allow Fire, Building, Parks, and other departments to understand and communicate detailed real property information. This impressive technology allows city departments to collaborate toward common goals by maximizing limited resources resulting in positive impact across Detroit.



# MOTOR CITY MAPPING

**Public data collected via Blexting and the MCM process will be made available to citizens, government agencies, and the community at large through an on-line platform.**

**FIRE FIGHTER:** Before arriving at the scene, fire fighters will know the property condition and if the structure is vacant or occupied. After a call, fire fighters will update property condition notifying relevant departments.

**BUSINESS OWNER:** Vacancies in commercial corridors can be identified by city planners and business owners, concentrating activity and reactivating vacant storefronts.

**COMMUNITY GROUP:** Neighborhood updates of property conditions can be carried out by community groups. City agencies can then be updated on problem properties via a common platform supported with parcel based data and photography.

**RESIDENT:** Residents can keep track of blight and the condition of their neighborhood, as well as provide observational information into the platform.

**PROSPECTIVE HOMEOWNER:** The MCM won't just identify problems, it can also reveal opportunities. Valuable property information can be evaluated by prospective purchasers and investors.

**POLICE OFFICER:** Illegal renting, scrapping, and other illegal activities can be reported through the MCM. The police will also be able to identify trends and frequency through the real-time visible date accessible in the MCM.

# TECHNOLOGY + DATA RECOMMENDATIONS

**RECOMMENDATION 3-1. Implement Motor City Mapping Phase II and Phase III.** Phase II of Motor City Mapping is the creation of a centralized, public database. This database will contain information from Motor City Mapping Phase I, plus data from a wide variety of sources, including numerous municipal departments. The project is planned for launch in spring 2014 and scheduled to be completed in fall 2014.

Phase II has four deliverables:

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**RECOMMENDATION 3-2. Annual survey of all 380,000 parcels.** Keeping the data and the MCM current is paramount to the viability of the entire system. The Task Force recommends that all 380,000 parcels in the City of Detroit be surveyed annually. The more effective way of surveying the entire city would be to allocate the parcels over a 12 month schedule requiring significantly fewer resources to accomplish the goal of annually updating information on each property. There is nothing that compels the simultaneous surveying of all properties during a short period of time. It would be very short sighted to not allocate the relatively inexpensive amount of resources needed to keep the information current.

**RECOMMENDATION 3-3. "Blexting Boot Camps".** The Task Force recommends conducting public training sessions (Blexting Boot Camps), led by DDD and Loveland Technologies, which would be integrated into regular community meetings attended by the city's new District Managers.

Participants will be trained to use the Public Dashboard and the Blexting application. The training will enable participants to read and interpret the Public Dashboard and contribute information to the MCM. Residents who do not own a computer or smart phone will always have the ability to work directly with their District Manager to obtain information from the Public Dashboard and provide information to the MCM.

These blexting boot camps should rotate between districts during the next three to six months. Special, additional boot camps should also be held for community leaders interested in becoming "Blexting captains." These leaders will be trained to conduct their own classes on Blexting and the Public Dashboard.

**RECOMMENDATION 3-4. Neighborhood Reports.** Phase II of MCM will allow District Managers to generate in depth parcel and neighborhood related activity reports. The Task Force recommends that reporting be communicated by the District Managers at neighborhood and community meetings on a monthly basis.

**RECOMMENDATION 3-5. Open Data Policy.** Blight is a shared problem among all city stakeholders: city agencies, organizations, private companies, and citizens. The Task Force strongly encourages all stakeholders to share and collaborate non-private information in a visible, open, and real-time environment.

It is imperative the city make a strategic investment in technology and create a philosophy of open and enthusiastic sharing of information among city employees, departments, and citizens.

This mission critical element must be supported from the city's top leadership on down to not only realize the success of the blight removal initiative but also to achieve a bright future for every neighborhood.